

Abdellah Fourtassi

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Research Profile. I combine theoretical and practical tools from computer science to build formal models of human cognition. I use machine learning as a model of cognitive development, with a particular focus on first language acquisition. I am especially interested in quantifying the role of multimodal and social grounding in learning.

APPOINTMENT

Aix Marseille University

Maître de Conférences HDR (~ Associate Professor), Computer Science, 2025–present
Maître de Conférences (~ Assistant Professor), Computer Science, 2019–2025
Researcher, Natural Language Processing research group (TALEP), 2019–present
Research fellow, Institute of Language, Communication & the Brain (ILCB), 2019–present
Co-Director, Master's program of Cognitive Science (MASCO), 2020–2021

Institut National de Recherche en Sciences et Technologies du numérique (INRIA, Paris)

Visiting Researcher, *Délégation Recherche INRIA*, 2021–2022

Stanford University

Postdoctoral Research Fellow, Department of Psychology, 2016–2019

EDUCATION

Ecole Normale Supérieure (ENS), Paris

Ph.D. in Cognitive Science, *summa cum laude*, 2015

Ecole des Hautes Etudes en Sciences Sociales (EHESS), Paris

Master's degree in Cognitive Science (CogMaster), 2012

Ecole Polytechnique, Paris

Science & Engineering diploma, 2012

Lycée Moulay Youssef, Rabat

Undergraduate studies in Mathematics & Physics, CPGE/MP*, 2006–2008

DISTINCTIONS & AWARDS

Keynote Speaker, *Child Language Symposium (CLS)*, 2026

Keynote Speaker, *SIGDIAL*, 2025

Aix Marseille University Best PhD Award, 2024 (thesis by M. Nikolaus; co-supervised with L. Prévot)

Marie Skłodowska-Curie European Fellowship, €185k, 2019 (declined; tenured position at AMU)

Stanford Data Science Seed Grant Award, \$50k, 2018

Fyssen Foundation's postdoctoral Study Grant, €40k, 2016

Visiting Scholar, Stanford University, March–July, 2016 (funded research position)

France-Stanford Center for Interdisciplinary Studies' Collaborative Project Award (\$15k), 2015–2016

Finalist, CoNLL Best Paper Award – *Fourtassi & Dupoux (2014)*

ACL Best Student Paper Award (\$1,000) – *Fourtassi & Dupoux (2013)*

Ranked 2nd/30, CMCL shared task on Eye-Tracking Prediction – *Dary, Nasr, & Fourtassi (2021)*

Doctoral Fellowship (AMX — Allocation-Moniteur Polytechnicien), 2012–2015

Eiffel Excellence Scholarship (undergraduate) — French Ministry of Foreign Affairs, 2008–2012

Early distinctions: Admission to École Polytechnique via competitive examination; ranked 12th nationwide in Morocco's national engineering-school entrance examination.

RESEARCH GRANTS

ANR JCJC (France), €287k (PI), 2022–2025

Mastering the Art of Conversation in Middle Childhood (MACoMiC)

ESRC (UK), £300k (co-PI, PI: G. Vigliocco, UCL), 2024–2027

Language Acquisition in the Early Years: Learning From and With Caregivers

Leverhulme Trust (UK), £290k (co-PI, PI: G. Vigliocco, UCL), 2025–2028

What Makes Social Interaction a Powerful Learning Tool

PEER-REVIEWED PUBLICATIONS

Book

Fourtassi, A. (under contract). *Neural Networks for a Theory of Language Development*.
Cambridge University Press

Edited Special Issue

Agee, L., Monfils, M, **Fourtassi, A.** (2025). Editorial: The Neural and Behavioral Methods of Social Learning. *Frontiers in Human Neuroscience*

Apidianaki, M., **Fourtassi, A.**, and Padó, S. (2024). Editorial: Language Learning, Representation, and Processing in Humans and Machines. *Computational Linguistics*

Journal papers

§ indicates student authors under my supervision

1. §Agrawal, A., Favre, B., & **Fourtassi, A.** (under revision). Mapping the Communicative Landscape of Early Child-Caregiver Dialogue.
2. **Fourtassi, A.**, & §Xu, Z. (under review). Initiating vs. Responding to Questions in Child-caregiver Spontaneous Conversations Across Preschool.
3. Nikolaus, M., & **Fourtassi, A.** (2026). Modeling Children's Grammar Learning via Caregiver Feedback in Natural Conversations. *Philosophical Transactions of the Royal Society B*, 381.
4. Bodur, K., **Fourtassi, A.**, Fredouille, C., Meunier, C. (2025). Speech Reduction in Middle Childhood: the Case of Child-Caregiver Interactions. *First Language*.
5. §Mazzocconi, C., O'Brien, B., §Bodur, K., & **Fourtassi, A.** (2025). Do Children Laugh Like Their Parents? Conversational Laughter Mimicry Occurrence and Acoustic Alignment in Middle-Childhood. *Journal of Nonverbal Behavior*, 49.
6. Agee, L., **Fourtassi, A.**, and Monfils, M. (2025). Social Context as a Source of Variability in the Psychological Sciences. *Frontiers in Behavioral Neuroscience*, 18.
7. §Cabiddu, F., §Nikolaus, M., & **Fourtassi, A.** (2025). Comparing Children and Large Language Models in Word Sense Disambiguation: Insights and Challenges. *Language Development Research*, 5(1).
8. §Bodur, K., §Nikolaus, N., Prévot, L., & **Fourtassi, A.** (2023). Using video calls to study children's conversational development: The case of backchannel signaling. *Frontiers in Computer Science*, 5.
9. §Wilson, K., Frank, M., C., & **Fourtassi, A.** (2023). Conceptual Hierarchy in Child-Directed Speech: Implicit Cues are More Reliable. *Journal of Cognition & Development*, 24(4).

10. §Nikolaus, M. & **Fourtassi, A.** (2023). Communicative Feedback in Language Acquisition. *New Ideas in Psychology*, 68, 100985.
11. §Nikolaus, M., §Maes, E., Auguste, J., Prévot, L., & **Fourtassi, A.** (2022). Large-scale Study of Speech Acts' Development in Early Childhood. *Language Development Research*, 2(1).
12. §Jiang, H., Frank, M., C., Kulkarni, V., & **Fourtassi, A.** (2022). Exploring Patterns of Stability and Change in Caregivers' Word Usage Across Early Childhood. *Cognitive Science*, 46 (7).
13. **Fourtassi, A.**, §Regan, S., & Frank, M., C. (2021). Continuous Developmental Change Explains Discontinuities in Word Learning. *Developmental Science*, 24(2).
14. **Fourtassi, A.** & Frank, M. C. (2020) How Optimal is Word Identification Under Multimodal Uncertainty? *Cognition*, 199.
15. **Fourtassi, A.**, §Bian, Y. & Frank, M. C (2020). The Growth of Children's Semantic and Phonological Networks: Insight from 10 Languages. *Cognitive Science*, 44 (7).

Peer-reviewed Proceedings

16. Jumelet, J., **Fourtassi, A.**, Haga, A., Bunzeck, B., Shandilya, B., Galvan-Sosa, D., Haznitrana, F. G., Padovani, F., Meyer, F., Hu, H., Etzaniz, J., Prevot, L., He, L., Grandury, M., Marcheva, M., Foroutan, N., Theodoropoulos, N., Sadeghi, P., Song, S., ... Choshen, L. (under review). *BabyBabelLM: A Multilingual Benchmark of Developmentally Plausible Training Data*.
17. §Brahimi, Y., §Blanc, C., & **Fourtassi, A.** (2026). Predicting Turn-Taking in Child-Adult Conversations Using Voice Activity Projection. In **IWSDS** (16th International Workshop on Spoken Dialogue Systems Technology).
18. §Liu, J., & **Fourtassi, A.** (2025). Benchmarking LLMs for Mimicking Child-Caregiver Language in Interaction. In **SemDial** (29th Workshop on the Semantics and Pragmatics of Dialogue)
19. §Agrawal, A., Favre, B., & **Fourtassi, A.** (2025). Identifying Repair Opportunities in Child-Caregiver Interactions. In **SemDial** (29th Workshop on the Semantics and Pragmatics of Dialogue)
20. §Agrawal, A., §Nikolaus, M., Favre, B., & **Fourtassi, A.** (2024). Automatic Coding of Contingency in Child-Caregiver Conversations. In **LREC-COLING** (The Joint International Conference on Computational Linguistics, Language Resources, and Evaluation)

21. §Nikolaus, M., §Agrawal, A., §Kaklamanis, P., Warstadt, A., & **Fourtassi, A.** (2024). Automatic Annotation of Grammaticality in Child-Caregiver Conversations. In **LREC-COLING** (The Joint International Conference on Computational Linguistics, Language Resources, and Evaluation)
22. §Goumri, D., §Agrawal, A., §Nikolaus, M., §VU, T., Bodur, K., §Semmar, E., §Armand, C., §Mazzocconi, C., Gupta, S., Prévot, L., Favre, B., Becerra, L., & **Fourtassi, A.** (2024). A Developmental Corpus of Child-Caregiver's Face-to-face vs. Video Call Conversations in Middle Childhood. In **LREC-COLING** (The Joint International Conference on Computational Linguistics, Language Resources, and Evaluation)
23. §Schweitzer, M., Montagnini, A., **Fourtassi, A.**, & Schatz, T. (2024). Multimodal Integration in Audio-Visual Speech Recognition: How Far Are We From Human-Level Robustness? In **NeurIPS** Workshop on Behavioral Machine Learning.
24. §Goumri, D.-E., Becerra, L., & **Fourtassi, A.** (2024). Child-Caregiver Gaze Dynamics in Naturalistic Face-to-Face Conversations. In **CogSci** (46th Annual Meeting of the Cognitive Science Society).
25. §Agrawal, A., Favre, B., & **Fourtassi, A.** (2024). Analyzing Communicative Intent Coordination in Child-Caregiver Interactions. In **CogSci** (46th Annual Meeting of the Cognitive Science Society).
26. §Peirola, M., § Xu, Z., & **Fourtassi, A.** (2024). Development of Flexible Role-Taking in Conversations Across Preschool. In **CogSci** (46th Annual Meeting of the Cognitive Science Society).
27. §Goumri, D., Janssoone, T., Becerra, L., & **Fourtassi, A.** (2023). Automatic Detection of Gaze and Smile in Children's Video Calls. In **ICMI Companion** (International Conference on Multimodal Interaction)
28. §Cabiddu, F., §Nikolaus, M., & **Fourtassi, A.** (2023). Modeling Children's Word Sense Disambiguation with Large Language Models: Insights and Challenges. In **CogSci** (45th Annual Meeting of the Cognitive Science Society).
29. §Nikolaus, M., Prévot, L., & **Fourtassi, A.** (2023). Communicative Feedback Signals in Response to Children's Grammatical Errors. In **CogSci** (45th Annual Meeting of the Cognitive Science Society).
30. §Mazzocconi, C., O'Brien, B., El Haddad, K., §Bodur, K., **Fourtassi, A.** (2023). Differences between Mimicking and Non-Mimicking Laughter in Child-Caregiver Conversation: A

Distributional and Acoustic Analysis. In **CogSci** (45th Annual Meeting of the Cognitive Science Society).

31. §Agrawal, A., §Liu, J., §Bodur, K., Favre, B., & **Fourtassi, A.** (2023). Development of Multimodal Turn Coordination in Conversations: Evidence for Adult-like Behavior in Middle Childhood. In **CogSci** (45th Annual Meeting of the Cognitive Science Society).
32. Tiran, T., Meewis, F., & **Fourtassi, A.**, & Dautriche I. (2023). Typology of Topological Relations Using Machine Translation. In **CogSci** (45th Annual Meeting of the Cognitive Science Society).
33. §Nikolaus, M., Salin, E., Ayache, S., **Fourtassi, A.**, & Favre, B. (2022). Do Vision-and-Language Transformers Learn Predicate-Noun Dependencies? In **EMNLP** (Conference on Empirical Methods in Natural Language Processing).
34. §Liu, J., §Nikolaus, M., §Bodur, K., & **Fourtassi, A.** (2022). Predicting Backchannel Signaling in Multimodal Child-Caregiver Conversation. In **ICMI Companion** (International Conference on Multimodal Interaction)
35. §Misiek, T. & **Fourtassi, A.** (2022). Caregivers Exaggerate Their Lexical Alignment to Young Children Across Several Cultures. In **SemDial** (26th Workshop on the Semantics and Pragmatics of Dialogue).
36. §Hallart*, C., §Peirola*, M., §Xu, Z., & **Fourtassi, A.** (2022). Development of Dialog Contingency in Child-Caregiver Early Interaction (* shared first authorship). In **SemDial** (26th Workshop on the Semantics and Pragmatics of Dialogue).
37. §Bodur, K., §Nikolaus, M., Prévot, L., & **Fourtassi, A.** (2022). Backchannel Behavior in Child-Caregiver Video Calls. In **CogSci** (44th Annual Meeting of the Cognitive Science Society).
38. §Nikolaus, M., Prévot, L., & **Fourtassi, A.** (2022). Communicative Feedback as a Mechanism Supporting the Production of Intelligible Speech in Early Childhood. In **CogSci** (44th Annual Meeting of the Cognitive Science Society).
39. §Nikolaus, M., & **Fourtassi, A.** (2021). Modeling the Interaction between Perception-based Learning and Production-based Learning in Children's Early Acquisition of Semantic Knowledge. In **CoNLL** (Conference on Computational Natural Language Learning)
40. §Dary, F., Nasr, A., & **Fourtassi, A.** (2021). TALEP at CMCL 2021 Shared Task: Non-Linear Combination of Low and High-level Features for Predicting Eye-Tracking Data. In **CMCL** (11th Workshop on Cognitive Modeling and Computational Linguistics).
2nd place (out of 30 submissions)

41. §Bodur, K., §Nikolaus, M., §Kassim, F., Prévot, L., & **Fourtassi, A.** (2021). ChiCo: A Multimodal Corpus for the Study of Child Conversation. In **ICMI Companion** (International Conference on Multimodal Interaction).
42. §Nikolaus, M. & **Fourtassi, A.** (2021). Evaluating the Acquisition of Semantic Knowledge from Cross-situational Learning in Artificial Neural Networks. In **CMCL** (The 11th Workshop on Cognitive Modeling and Computational Linguistics).
43. §Nikolaus, M., §Maes, E., Auguste, J., Prévot, L., & **Fourtassi, A.** (2021). Large-scale Study of Speech Acts' Development Using Automatic Labeling. In **CogSci** (43th Annual Meeting of the Cognitive Science Society).
44. §Nikolaus, M., §Maes E., & **Fourtassi, A.** (2021). Modeling Speech Act Development in Early Childhood: The Role of Frequency and Linguistic Cues. In **CogSci** (43th Annual Meeting of the Cognitive Science Society).
45. §Dary, F., **Fourtassi, A.**, & Nasr, A., (2021). On the Role of Low-level Linguistic Tasks for Reading Time Prediction. In **CogSci** (43th Annual Meeting of the Cognitive Science Society).
46. §Misiek, T., Favre, B., & **Fourtassi, A.** (2020). Development of Multi-level Linguistic Alignment in Child-Adult Conversations. In **CMCL** (The 11th Workshop on Cognitive Modeling and Computational Linguistics)
47. **Fourtassi, A.** (2020). Word Co-occurrence in Child-directed Speech Predicts Children's Free Word Associations. In **CMCL** (The 11th Workshop on Cognitive Modeling and Computational Linguistics).
48. **Fourtassi, A.**, §Wilson, K., & Frank, M., C. (2020). Discovering Conceptual Hierarchy Through Explicit and Implicit Cues in Child-Directed Speech. In **CogSci** (42nd Annual Meeting of the Cognitive Science Society).
49. **Fourtassi, A.**, §Scheinfeld, I., & Frank, M. C.(2019) The Development of Abstract Concepts in the Children's Early Lexical Networks. In **CMCL** (The 10th Workshop on Cognitive Modeling and Computational Linguistics).
50. **Fourtassi, A.**, §Regan, S. & Frank, M. C. (2019). The Role of Continuous Developmental Change in Explaining Discontinuities in Word Learning. In **CogSci** (41st Annual Meeting of the Cognitive Science Society).
51. **Fourtassi, A.**, & Dupoux, E. (2019) Phoneme Learning is Influenced by the Taxonomic Similarity of the Semantic Referents. In **CogSci** (41st Annual Meeting of the Cognitive Science Society).

52. **Fourtassi, A.**, §Bian, Y., & Frank, M., C. (2018). Word Learning as Network Growth: A Cross-Linguistic Analysis. In **CogSci** (40th Annual Meeting of the Cognitive Science Society).
53. **Fourtassi, A.** & Frank, M., C. (2017). Word Identification Under Multimodal Uncertainty. In **CogSci** (39th Annual Meeting of the Cognitive Science Society).
54. **Fourtassi, A.** & Dupoux, E. (2016). The role of word-word co-occurrence in word meaning learning. In **CogSci** (38th Annual Meeting of the Cognitive Science Society).
55. **Fourtassi, A.** & Dupoux, E. (2014). Rudimentary Lexicon and Semantics Help Bootstrap Phoneme Acquisition. In **CoNLL** (Conference on Computational Natural Language Learning)
Best paper award finalist
56. **Fourtassi, A.**, Schatz, T., Varadarajan, B. & Dupoux, E. (2014). Exploring the Relative Role of Bottom-Up and Top-Down Information in Phoneme Learning. In **ACL** (52nd Annual Meeting of the Association for Computational Linguistics).
57. **Fourtassi, A.**, Dunbar, E., & Dupoux, E. (2014). Self Consistency as an Inductive Bias in Early Language Acquisition. In **CogSci** (36th Annual Meeting of the Cognitive Science Society).
58. **Fourtassi, A.**, Borschinger, B., Johnson, M. & Dupoux, E. (2013). Why is English so easy to segment? In **CMCL** (The 4th Workshop on Cognitive Modeling and Computational Linguistics).
59. **Fourtassi, A.** & Dupoux, E. (2013). A Corpus-based Evaluation Method for Distributional Semantic Models. In **ACL** student research workshop
IBM Best Student paper (\$1,000)
60. Jansen, A., Dupoux, E., Goldwater, S., Johnson, M., Khudanpur, S., Church, K., Feldman, N., Hermansky, H., Metze, F., Rose, R., Seltzer, M., Clark, P., McGraw, I., Varadarajan, B., Bennett, E., Borschinger, B., Chiu, J., Dunbar, E., **Fourtassi, A.**, Harwath, D., Lee, C., Levin, K., Norouzian, A., Peddinti, V., Richardson, R., Schatz, T. & Thomas, S. (2013). A summary of the 2012 JHU CLSP workshop on zero resource speech technologies and models of early language acquisition. In **ICASSP** (The International Conference on Acoustics, Speech and Signal Processing)

INVITED TALKS

Keynote Lectures

Child Language Symposium (CLS), *Deep Learning for a Theory of Language Development*, 2026

SIGDIAL, *The Primary Role of Dialogue in Language Development*, 2025

TALN Workshop on AI for Education, *Can Large Language Models Learn Like Children?* 2025

Invited Talks — Workshops & Expert Meetings

Machine Learning to Study Home Audio Recordings, Online Event, International Congress of Infant Studies (ICIS), June 19, 2025 ([link](#))

State of the Art in Scientific Psychology, National Meeting, Comité National Français de Psychologie Scientifique (CNFPS), Oct 25, 2025 ([link](#))

NLP in the Era of Generative AI, Cognitive Sciences, and Societal Transformation, Workshop, Mila Quebec Artificial Intelligence Institute, Oct 1–Oct 3, 2024 ([link](#))

Developmental Machine Learning: From Human Learning to Machines and Back, Dagstuhl Seminar, Oct 16–Oct 21, 2022 ([link](#))

AI and the Productivity of Science, Workshop, Organisation for Economic Co-operation and Development (OECD), Oct 29–Nov 5, 2021 ([link](#))

Invited Talks — Universities & Research Consortia (Selection)

Social Contingency Consortium, 2023, 2025

Cambridge University, Center for Family Research, 2024

University College London, Psychology and Language Sciences, 2023, 2024

International Centre for Language and Communicative Development (LuCiD), 2023

University of Göttingen, Psychology of Language Department, 2023

The Chinese University of Hong Kong, Department of Linguistics, 2022

INRIA Lille Nord Europe, MGNET Seminar, 2022

Radboud University, Multimodal Language and Cognition Seminar, 2022

Pr[AI]rie institute Colloquium, 2022

University of Aarhus, Interacting Minds Center Seminar, 2022

University of Amsterdam, Computational Linguistics Seminar (CSL), 2022

University Grenoble Alpes, GIPSA-lab seminar, 2022

Stanford University, Center for the Study of Language & Information (CSLI), 2018

MENTORSHIP

Ph.D. Supervision — Primary Role

Marianne Schweitzer (2025–present) — co-advised with C. François

Funding: Amidex fellowship

Youcef Brahim (2025–present)

Funding: ED184 fellowship

Dhia Elhak Goumri (2022–present) — co-advised with L. Becerra

Funding: MACoMiC grant (ANR JCJC)

Defense expected: March 2026

Abhishek Agrawal (2022–2025) — co-advised with B. Favre

Funding: MACoMiC grant (ANR JCJC)

Defended: October 2025

Mitja Nikolaus (2020–2023) — co-advised with L. Prévot

Funding: ILCB fellowship

*Defended: September 2023; **Aix Marseille University Best PhD Award***

Current position: Postdoctoral Researcher

Ph.D. Supervision — Secondary Role

Mathias Vigouroux (2025–present) — co-advised with J. Sackur

Funding: ED540

Franck Dary (2018–2022) — co-advised with A. Nasr

Funding: ED184 fellowship

Defended: July 2022

Current position: Data Scientist Tech Lead, DialOnce

Postdoctoral Researchers

Giulia Rambelli (2026–present) — co-advised with L. Prévot

Funding: ILCB Fellowship

Monica Barbir (2024–present)

Funding: Marie Skłodowska-Curie European Fellowship

Chiara Mazzocchi (2020–2023) — co-advised with B. Priego-Valverde

Funding: ILCB Fellowship

Master's Interns

* Supervision resulted in peer-reviewed publication

Youcef Brahim* — Image and Signal Processing (M2), Toulouse University, 2025
 César Blanc* — Artificial Intelligence (M2), Aix Marseille University, 2025
 Nina Gregorio — Cognitive Science (M1), Ecole Normale Supérieure, 2023
 Jing Liu* — Linguistics & Communication (M2), Radboud University, 2022
 Maryam Taiaa — Data Science (M2), Aix Marseille University, 2022
 Laudine LÊ — Biometrics (M2), Paris-Est University, 2022
 Inès Ben Haj Kacem — Cognitive Science (M1), Ecole Normale Supérieure, 2022
 Sarah Said — Data Science (M1), Paris-Dauphine University, 2021
 Gabriel Meunier — Artificial Intelligence (M2), Aix Marseille University, 2021
 Kübra Bodur* — Cognitive Science (M2), Aix Marseille University, 2021
 Maureen Astier — Cognitive Science (M1), Aix Marseille University, 2021
 Pelayo Mencos — Cognitive Science (M2), Aix Marseille University, 2020
 Thomas Misiek* — Cognitive Science, Aix Marseille University, 2020
 Hang Jiang* — Symbolic Systems (M1-M2), Stanford University, 2019

Summer Interns

Martin Cuingnet — Ecole Normale Supérieure Paris-Saclay
 Lucien Loriferne — Ecole Normale Supérieure Lyon
 Petros Kaklamanis* — University of Pennsylvania
 Martin Padovani — Ecole Polytech' Marseille
 Hong Duc Thang Vu* — Aix Marseille University
 Elias Semmar* — Aix Marseille University
 Cassandre Armand* — Aix Marseille University
 Charlie Hallart* — Aix Marseille University
 Morgane Peirollo* — Aix Marseille University
 Eliot Maes* — Aix Marseille University
 Yassine Laribi — Ecole Centrale Marseille
 Zihan Xu* — Jiao Tong University
 Kyra Wilson* — Stanford University
 Yifan Ma — Stanford University
 Sophie Regan* — Stanford University
 Isaac Scheinfeld* — Stanford University

Yuan Bian* — Stanford University

Thesis Advisory Committee

Khazar Khorrami (Tampere University), May Abdou (Ecole Normale Supérieure), Kaijia Tey (Ecole Normale Supérieure), Rozic Gal (University College London), Rudradeep Guha (Université de Bourgogne), Daria Goryachun (Aix Marseille University), Maïwenn Fleig (Aix Marseille University), Ambre Balleroy (Aix Marseille University), Leonardo Pinto Arata (Aix Marseille University), Auriane Boudin (Aix Marseille University), Alafate Abulimiti (INRIA Paris), Léo Jacqmin (Aix Marseille University), Salma Ferjani (Aix Marseille University).

TEACHING

Aix Marseille University

Selected courses (Design & Instruction)

Building Minds: the Programming Tools—2025–present

Master in Cognitive Science (MASCO)—mandatory courses.

Description: Introduces formal cognitive modeling through programming: students use Python to build a cognitive agent from scratch (perception–action cycles) and assess how adding memory, planning, learning, and communication improves adaptive behavior in simulated environments.

Machine Learning for Cognitive Science—2024–present (with T. Schatz)

Master in Cognitive Science (MASCO)—mandatory courses.

Master in Computer Science (IAAA)—optional courses.

Description: Introduces machine-learning methods as tools for cognitive modeling, showing how accounts of (neuro)cognitive processes can be linked to real-world behavior in ecological settings. Hands-on case studies in perception and learning using auditory, visual, and social data.

Other courses (selection)

Data Science (L1, Computer Science), 2025

Introduction to Machine Learning (L3, Computer Science), 2022–2023

Natural Language Processing (M1-IAAA, Computer Science), 2020–2023

Probability Theory for Computer Science (M1-IAAA, Computer Science), 2022

Theories in Cognitive Science (M1-MASCO, Cognitive Science), 2020–2022

Summer Schools

ILCB summer school

Deep Learning for a theory of Language Development, 2026 (week-long course)

AI for the Ecological Study of Language Development, 2025 (week-long course)

Child language acquisition: State of the art, 2021–2024 (single-lecture course)

ETAL (National Summer School on NLP)

When NLP meets Cognitive Science, 2023 (single-lecture course)

SERVICE

Editorial Service

Conference editorial roles

Program Chair, ACL Workshop *CDL* — 2026

Senior Area Chair, COLING — 2025

Area Chair, ACL Rolling Review — 2025

Area Chair, ACL — 2023

Journal editorial roles

Associate Editor, *Frontiers in Cognitive Neuroscience* — 2023–present

Guest Editor, *Computational Linguistics* — 2024

Special issue: *Language Learning, Representation, and Processing in Humans and Machines*

Co-editors: M. Apidianaki (UPenn), S. Padó (University of Stuttgart)

Guest Editor, *Frontiers in Human Neuroscience* — 2025

Special issue: *Neural and Behavioral Mechanisms of Social Learning*

Co-editors: Laura Agee (UT Austin), Marie H. Monfils (UT Austin)

Reviewing Service

Funding Agencies

Agence Nationale de la Recherche (ANR, France)

National Science Foundation (NSF, USA),
 The British Academy (UK)
 The Royal Society (UK)
 IVADO AI (Canada)
 National Science Center (Poland)

Book Proposals

Cambridge University Press

Journals (selection)

Psychological Review, Trends in Cognitive Science, Proceedings of The Royal Society B, Cognition, Cognitive Science, Cognitive Psychology, Child Development, Journal of Experimental Psychology: General, Developmental Psychology, Developmental Cognitive Neuroscience, Open Mind: Discoveries in Cognitive Science, Speech Communication, Language Learning, Infant Behavior & Development, Language Learning & Development, Frontiers Human Neuroscience, Behavior Research Methods, Language & Cognition.

Conferences (selection)

ACL Rolling Review (ARR), Cognitive Science Society Conference (CogSci), International Conference on Multimodal Interaction (ICMI), Cognitive Development Society Conference (CDS), Conference on Computational Natural Language Learning (CoNLL), International Conference on Affective Computing and Intelligent Interaction (ACII), Workshop on Cognitive Modeling and Computational Linguistics (CMCL), Workshop on Knowledge Extraction and Integration for Deep Learning (DeeLIO).

Governance & Institutional Service

Governing Board, Institute of Language Communication & the Brain (ILCB), 2020–present

Co-Founder and Director (with L. Prévot), CoCoDev project-team, ILCB, 2020–present

Co-Founder and Coordinator (with P. Denis and L. Prévot), National Working Group (GT)

“*NLP & Cognition*” GDR TAL (CNRS), 2025 –present

Co-Director (with B. Isableu), Master in Cognitive Science (MASCO), 2020–2021

Selection Committee, Maître de Conférences position in Computer Science, Univ. Lille, 2025

Organizer, The CoCoDev Seminars (archive and recordings), 2020–present

Event co-organization

Workshop on *Computational Developmental Linguistics* (CDL), ACL, 2026

National Meeting of the GT “*NLP & Cognition*” GDR TAL (CNRS), 2025

Workshop on Multimodality in Social Interactions (MSI), 2024

National Summer School on NLP (ETAL), 2023

National meeting *Bridging NLP and Cognitive Science*, GDR TAL (CNRS), 2020

Media & Public Outreach

Interactive stand, *CoCoDev* research, Festival Explore, 2026

Interview with Photizo on IA (Youtube podcast): *IA and Cognitive Science*, 2025

Interactive stand, *the MACOMIC ANR project*, Nuit des chercheurs, 2023

Public outreach article, “Les bébés apprennent l’art de la conversation avant même de savoir parler” in *The Conversation* (link), 2022